

Industrial Internship Report on

"Weed Detection and Smart City Traffic Pattern Forecasting — A Machine Learning & Data Analytics Project Suite"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project suite provided by UCT, to be fully implemented and documented within a six-week timeline. The objective was to apply Python programming, machine learning, and data analytics to solve real-world problems in the domains of precision agriculture and urban traffic management.

My project involved designing and developing two independent, industry-relevant applications: a Weed Detection system that uses image processing and machine learning to distinguish weeds from crop plants, and a Smart City Traffic Pattern Forecasting system that analyses historical traffic data and predicts congestion trends to support smarter urban planning. Each application addressed a distinct real-world challenge — one in precision agriculture and one in intelligent transportation — while sharing a common foundation in Python-based data handling and model building.

This internship gave me exceptional exposure to industry-relevant engineering challenges. Implementing these two applications from scratch — each with distinct data pipelines, modelling approaches, and evaluation methods — sharpened my ability to write clean, modular, and maintainable Python code, and deepened my understanding of applied machine learning. It was a deeply enriching experience that bridged academic learning with real-world software development practice.

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1. Preface

Over the course of six weeks, I systematically worked through the full development lifecycle for both assigned projects. Week 1 was dedicated to understanding the problem statements, setting up the development environment, sourcing sample datasets, and beginning exploratory work on the Weed Detection image dataset and the traffic dataset for the Smart City project.

Week 2 involved in-depth study through the assigned e-book "Learning Python" by Mark Lutz and a supplementary video tutorial, followed by Pre-Quiz-1. Week 3 focused on hands-on learning of NumPy and Pandas through Jupyter Notebook exercises, which formed the analytical foundation for both projects. Week 4 involved completing python_Pre: Quiz-2 as a formal knowledge assessment. Weeks 5 and 6 were dedicated to building and evaluating the Weed Detection classification model and the Traffic Pattern forecasting model, refining all scripts, conducting testing, and preparing the final project repository and this report.

Practical internships are indispensable in the career development of software and data engineering students. They provide the environment to apply theoretical knowledge to tangible problems, build professional coding habits, and develop the analytical mindset required in the industry. This internship fulfilled that need by presenting clearly defined, industry-relevant problem statements that demanded thoughtful engineering and modelling solutions.

The assigned project suite addressed two distinct real-world challenges: the difficulty of distinguishing unwanted weeds from crop plants at scale without excessive manual labour or indiscriminate herbicide use, and the lack of accessible, data-driven tools for understanding and anticipating traffic congestion patterns in growing urban areas. Each project was scoped to be a focused, standalone Python application solving one of these problems effectively.

I am sincerely grateful to upskill Campus (USC), The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT) for designing and delivering this structured internship program. The platform, resources, and mentorship provided enabled a focused, progressive, and industry-aligned learning experience. This opportunity has had a meaningful impact on my technical growth and professional confidence.

The program was structured as a six-week engagement: Week 1: Project orientation, environment setup, dataset collection, and initial exploration for the Weed Detection and Traffic Forecasting projects. Week 2: E-book study, video tutorial, and Pre-Quiz-1 assessment. Week 3: NumPy and Pandas modules with Jupyter Notebook practicals. Week 4: python_Pre: Quiz-2 assessment. Weeks 5-6: Model building, evaluation, and completion of both projects, full testing, GitHub deployment, and report preparation.

This internship provided an immersive, hands-on learning experience that significantly advanced my Python and applied machine learning skills. Working across two distinct project domains exposed me to a wide range of concepts — from image pre-processing and classification to time-series analysis and forecasting. The structured curriculum, combined with practical project work and formal assessments, made this one of the most productive and professionally rewarding learning experiences of my academic career.

I would like to express my sincere gratitude to the entire team at upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd for their effort in creating this well-structured internship program. I also thank the online developer community, open-source contributors, and the authors of the study materials that supported my learning throughout these six weeks.

To my fellow students and future interns: embrace project-based learning with full commitment. The process of building something from scratch — cleaning data, refining models, and seeing it finally work — is where real engineering growth happens. Stay consistent, practise daily, and always document your work. Your code tells your story; make it worth reading.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.

i. UCT IoT Platform (UCT Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications, at the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to:

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application (Power BI, SAP, ERP)
- Rule Engine

ii. Smart Factory Platform (Factory Watch)

Factory watch is a platform for smart factory needs. It provides Users/ Factory:

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service they want to start with and then scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solutions in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solutions leveraging Embedded systems, Industrial IoT and Machine Learning Technologies by finding the Remaining Useful Life of various Machines used in the production process.

2.2 About upskill Campus (USC)

upskill Campus, along with The IoT Academy and in association with Uniconverge Technologies, has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way, spanning career growth/upskilling, professional networking, a collaboration platform, and a job/internship platform.

2.3 The IoT Academy

The IoT academy is the EdTech Division of UCT that runs long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to:

- get practical experience of working in the industry.
- solve real world problems.
- have improved job prospects.
- have improved understanding of our field and its applications.
- have personal growth like better communication and problem solving.

2.5 Reference

[1] upskill Campus Official Website: <https://upskillcampus.com>

[2] UniConverge Technologies Pvt Ltd (UCT): <https://uniconverge.com>

[3] Mark Lutz, "Learning Python", 4th Edition, O'Reilly Media, 2009

[4] Jake VanderPlas, "Python Data Science Handbook", 1st Edition, O'Reilly Media, 2016

2.6 Glossary

Terms	Acronym
USC	upskill Campus - career development platform and internship organiser
UCT	UniConverge Technologies Pvt Ltd - industrial partner and project provider
IoT	Internet of Things - interconnected network of smart devices
API	Application Programming Interface - protocol for software communication
ML	Machine Learning - algorithms that learn patterns from data
CNN	Convolutional Neural Network - deep learning model used for image classification
ITS	Intelligent Transportation System - technology-driven traffic management systems

3. Problem Statement

In the assigned problem statement:

In today's digitally driven environment, agriculture and urban infrastructure face recurring challenges that, while addressed piecemeal in the past, are increasingly being solved through data and machine learning.

In precision agriculture, weeds compete with crops for nutrients, water, sunlight, and space, directly reducing yield and crop quality. Manual weeding is labour-intensive, slow, and does not scale to large farms, while blanket application of herbicide across an entire field is costly, environmentally harmful, and often unnecessary. There is a growing need for an automated system that can visually distinguish weeds from crop plants so that treatment can be targeted only where it is needed.

In urban transportation, rapidly growing city populations and vehicle density have made traffic congestion a persistent and costly problem, affecting commute times, fuel consumption, and air quality. Traffic authorities often lack lightweight, data-driven tools that can analyse historical traffic volume patterns and forecast near-term congestion trends at specific locations and times, which limits proactive planning such as adaptive signal timing or targeted infrastructure investment.

This project set was assigned to address both of these challenges through well-designed, independently functional Python applications: a Weed Detection system built on image processing and machine learning, and a Smart City Traffic Pattern Forecasting system built on time-series data analysis and predictive modelling.

4. Existing and Proposed solution

Existing approaches to these challenges are either manual, resource-intensive, or lack accessibility for small and mid-sized operators. Weed management is typically performed manually or through uniform herbicide spraying across an entire field, both of which are inefficient — manual weeding does not scale, while blanket spraying wastes chemicals and harms non-target areas of the field. Commercial precision-agriculture drones and vision systems exist, but they are expensive, proprietary, and often require specialised hardware and internet-connected cloud services. Traffic pattern analysis, similarly, is often handled by large, proprietary Intelligent Transportation System (ITS) platforms that require significant infrastructure investment, or is not performed at all in small and mid-sized cities, leaving traffic authorities to rely on historical intuition rather than data-driven forecasts. The fundamental limitation across both existing approaches is the absence of a lightweight, transparent, and Python-programmable solution that a student, researcher, or small organisation can build, understand, and extend.

The proposed solution is a two-part Python Application Suite that addresses each of the above challenges through a dedicated, standalone project:

- **Weed Detection Project:** A Python-based image classification pipeline that pre-processes leaf/field images using OpenCV (resizing, colour-space conversion, and noise reduction), extracts relevant visual features, and classifies each image as "Crop" or "Weed" using a trained machine learning model (a Convolutional Neural Network built with TensorFlow/Keras, benchmarked against a classical model built with scikit-learn). The output supports targeted herbicide application instead of blanket spraying.
- **Smart City Traffic Pattern Forecasting Project:** A Python-based data analytics pipeline that loads historical traffic volume data using Pandas, performs exploratory data analysis and visualisation (using Matplotlib/Seaborn) to identify peak-hour and location-wise congestion patterns, and applies a forecasting model (time-series analysis using statsmodels/ARIMA, benchmarked against a regression-based model in scikit-learn) to predict short-term traffic volume trends at a given junction or corridor.

The key value addition of this suite lies in its simplicity, transparency, and full reproducibility. Unlike proprietary commercial platforms, these projects require no specialised hardware, no third-party licensing, and no vendor lock-in. They are open-source, easily extensible, and serve as strong educational references for anyone learning applied machine learning and data analytics. The suite demonstrates that practical, production-relevant tools can be built with core Python, NumPy, Pandas, and open-source ML libraries alone — making it both a functional proof of concept and a powerful portfolio asset.

4.1 Code submission (GitHub link)

<https://github.com/ayushisbca2024-cell/upskillcampus>

4.2 Report submission (GitHub link)

https://github.com/ayushisbca2024-cell/upskillcampus/blob/main/WeedDetection_TrafficForecasting_Ayushi_USC_UCT.pdf

5. Proposed Design/ Model

The project suite follows a modular, pipeline-based architectural design. Each application is structured around clearly defined stages responsible for individual operations such as data ingestion, pre-processing, model training/inference, and output generation. This separation of concerns ensures clean code organisation and ease of maintenance.

Both applications leverage the following shared design principles:

- **Modular Pipeline Architecture:** Each stage is encapsulated in a dedicated function (e.g., `preprocess_image()`, `train_model()`, `predict_class()`, `load_traffic_data()`, `forecast_volume()`), promoting reusability and readability.
- **File-Based Data Persistence:** The Weed Detection project stores the trained classification model and labelled image datasets on disk for repeatable inference. The Traffic Forecasting project reads historical traffic data from a structured CSV dataset, enabling analysis updates without modifying the core script.
- **Procedural Flow Control:** Each project follows a clear main execution flow — initialise, load data, pre-process, train/infer, evaluate, and return output.
- **Error Handling:** Input validation and exception handling are implemented across both projects to ensure graceful failure and user-friendly feedback, including checks for missing files, corrupt images, and malformed data rows.

The design prioritises reproducibility, interpretability of results, and straightforward extensibility for future enhancements.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Weed Detection — Data Flow: Input Field/Leaf Image → Pre-process (Resize, Colour Conversion, Noise Removal via OpenCV) → Feature Extraction → Load Trained Model → Predict Class (Crop/Weed) → Confidence Score → Display/Log Result → [Optional] Trigger Targeted Action.

Traffic Pattern Forecasting — Data Flow: Load Historical Traffic Dataset (CSV via Pandas) → Clean & Structure Data → Exploratory Analysis (Peak Hours, Location-wise Volume) → Visualise Patterns (Matplotlib/Seaborn) → Train Forecasting Model (ARIMA/Regression) → Generate Short-Term Forecast → Evaluate Accuracy → Display Forecast & Trend Report.

6. Performance Test

The performance of each application in the suite was evaluated against real-world engineering and modelling constraints to ensure reliability, accuracy, and robustness beyond an academic prototype.

The following engineering constraints were identified and addressed for each application:

Weed Detection Project:

- **Constraint — Image Quality Variance:** Field images vary in lighting, angle, and background clutter. Resolved by applying consistent pre-processing (resizing, normalisation, and colour-space standardisation) before feeding images to the model.
- **Constraint — Class Imbalance:** Crop images significantly outnumbered weed images in the raw dataset. Resolved by applying data augmentation and class-weighting during model training.

Smart City Traffic Pattern Forecasting Project:

- **Constraint — Missing/Noisy Data:** Real-world traffic logs contained missing timestamps and outlier readings. Resolved by implementing data cleaning, interpolation, and outlier-capping before analysis.
- **Constraint — Forecast Drift:** Simple models lost accuracy over longer prediction horizons. Resolved by limiting forecasts to a validated short-term window and retraining periodically on recent data.

Test Results:

- **Weed Detection:** The classification model was evaluated on a held-out test set of labelled field images and achieved consistent accuracy in distinguishing crop from weed classes. Misclassifications were concentrated in visually ambiguous, overlapping-canopy images, which are noted as an area for future improvement.
- **Traffic Pattern Forecasting:** The forecasting model was evaluated against a held-out portion of historical traffic data using standard error metrics (MAE/RMSE) and produced forecasts that tracked observed peak-hour and off-peak trends closely across the test window.

Engineering constraint dimensions assessed: data quality (cleaning and validation applied to both image and tabular datasets), model performance (accuracy, precision/recall for classification; MAE/RMSE for forecasting), execution efficiency (both pipelines ran within standard desktop hardware and memory limits), and robustness (both scripts handled missing files, corrupt records, and malformed inputs without unhandled exceptions).

All identified constraints were addressed and verified through testing. For future production deployment, the following recommendations are noted: expand the weed image dataset with additional crop species and field conditions to improve generalisation; deploy the traffic forecasting model against a live data feed (e.g., IoT traffic sensors) for real-time predictions; and introduce a monitoring dashboard for both models to track performance drift over time.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7. My learnings

This six-week internship has been one of the most technically comprehensive and professionally formative experiences of my academic journey. The following key learnings were acquired:

- **Core Python Mastery:** Deepened my understanding of Python syntax, data types, control structures, functions, modules, and object-oriented principles through real project implementation.
- **Image Processing Fundamentals:** Gained hands-on experience with OpenCV for image loading, resizing, colour-space conversion, and noise reduction — foundational skills for any computer vision application.
- **Applied Machine Learning:** Learnt to build, train, and evaluate classification models (CNN-based and classical scikit-learn models) for the Weed Detection project, including handling class imbalance and avoiding overfitting.
- **Time-Series Analysis and Forecasting:** Learnt to structure, visualise, and forecast time-based traffic data using statsmodels/ARIMA and regression techniques, understanding key evaluation metrics such as MAE and RMSE.
- **NumPy and Pandas:** Completed structured learning in numerical computing and data manipulation using NumPy arrays and Pandas DataFrames through Jupyter Notebook practicals, establishing a foundation for data analysis and data science work.
- **Data Cleaning and Preparation:** Developed practical experience handling missing values, outliers, and inconsistent records in both image and tabular datasets before modelling.
- **Debugging and Problem Solving:** Developed systematic debugging skills through iterative testing of each project, learning to isolate errors, trace logic flows, and implement clean fixes.
- **Professional Documentation:** Learnt to document code, write structured reports, and prepare a GitHub repository for project submission — skills that are essential in any professional software engineering or data science role.

These competencies collectively position me strongly for roles in machine learning engineering, data analysis, computer vision, and applied data science.

8. Future work scope

The following professional enhancements are planned as future extensions of this project suite:

- **Weed Detection — Drone Integration:** Extend the image classification pipeline to process live video feeds from agricultural drones, enabling real-time, field-wide weed mapping.
- **Weed Detection — Model Upgrade:** Experiment with transfer learning using pretrained networks (e.g., MobileNetV2, ResNet) to improve accuracy and reduce training time on larger, more diverse datasets.
- **Weed Detection — Automated Spraying Integration:** Connect the classification output to a GPS-tagged, targeted spraying mechanism to fully automate precision herbicide application.
- **Traffic Forecasting — Real-Time Data Feed:** Integrate live IoT traffic sensor data or public traffic APIs to replace static historical datasets with continuously updating input.
- **Traffic Forecasting — Advanced Models:** Explore deep learning-based sequence models (e.g., LSTM) for improved long-horizon forecasting accuracy compared to classical time-series methods.
- **Web Deployment:** Deploy both projects as interactive web applications using Flask or Streamlit, allowing users to upload images or query traffic forecasts through a simple browser interface.
- **GitHub Portfolio and CI/CD:** Establish a professional GitHub repository with full documentation, README files, inline code comments, and automated testing pipelines using GitHub Actions to demonstrate industry-standard software delivery practices.
- **Dashboard and Visualisation Extension:** Build a unified dashboard (using Plotly Dash or Power BI) to visualise weed-detection statistics and traffic-forecast trends side by side for stakeholder reporting.